

Improving Hallucination Detection via Question-Aware Knowledge Summarization

Abdelrahman Mohamed

Computer Science Department
October University for Modern
Sciences and Arts (MSA)
Giza, Egypt

Abdelrahman.mohamed101@msa.edu.eg

Belal Hesham

Computer Science Department
October University for Modern
Sciences and Arts (MSA)
Giza, Egypt

belal.hesham@msa.edu.eg

Ebtesam E. Shemis

Computer Science Department
October University for Modern
Sciences and Arts (MSA)
Giza, Egypt

eelhosseiny@msa.edu.eg

Abstract—Hallucination detection in knowledge-grounded question answering requires determining whether a candidate answer is supported by the provided evidence or hallucinated. This task becomes more difficult when the knowledge passage contains long or irrelevant context that obscures the evidence needed for verification.

This paper investigates whether question-aware evidence compression can improve hallucination detection by increasing the signal-to-noise ratio of the verifier input. We propose a lightweight supervised verifier based on fine-tuned DeBERTa-v3-base, operating directly on evidence-question-answer triples (K, Q, A) . The verifier is used with an optional question-aware summarization preprocessing stage, where the knowledge passage is summarized conditioned on the question before verification.

The main benefit appears under distribution shift. On an unseen HotpotQA-derived binary verification benchmark with long and reasoning-intensive passages, question-aware FLAN-T5 summarization improves macro-F1 from near-chance performance (0.5040) to 0.7905. A length-matched truncation control reaches only 0.5457 macro-F1, indicating that the gain is not explained by input shortening alone. On HaluEval-QA, which is used as a controlled in-distribution benchmark, the best configuration achieves an F1 score of 0.9849.

These results suggest that question-aware evidence compression is especially useful for long-context hallucination verification. At the same time, the summarized setting should be viewed as a practical tradeoff: it improves robustness and verifier-side efficiency, while adding preprocessing cost when summaries are generated online.

Index Terms—Hallucination Detection, Knowledge-Grounded QA, DeBERTa, Supervised Learning, Question-Aware Knowledge Summarization, HaluEval

I. INTRODUCTION

Large language models (LLMs) can produce fluent answers that are not supported by the provided evidence. In knowledge-grounded question answering (QA), this is risky because users expect answers to be justified by an explicit knowledge passage rather than by the model’s parametric memory. Prior work also shows that extracting and organizing information from data is important for knowledge utilization and decision making [1].

We formulate hallucination detection as supervised verification over a knowledge passage K , question Q , and candidate answer A . The verifier predicts whether A is supported by K or hallucinated. Since knowledge passages may contain long

irrelevant context, we study whether question-aware summarization can increase the signal-to-noise ratio before verification. HaluEval-QA is used as a controlled in-distribution benchmark, while a HotpotQA-derived benchmark is used as a long-context out-of-distribution stress test. The novelty is therefore not a new backbone or summarizer, but a controlled study isolating the effect of question-focused evidence compression through fixed-verifier comparisons, leakage checks, length-matched controls, and external evaluation.

A. Contributions

The main contributions are: (i) a controlled empirical analysis of question-aware evidence compression for hallucination detection; (ii) a lightweight fine-tuned DeBERTa-v3 verifier operating on (K, Q, A) triples under black-box deployment constraints; (iii) evidence that question-aware compression improves accuracy, verifier-side efficiency, and long-context generalization; and (iv) comparisons with generic summarization, alternative backbones, factual-consistency baselines, and length-matched truncation.

II. RELATED WORK

HaluEval provides a controlled hallucination benchmark across tasks, including QA, where each instance contains a knowledge passage, question, grounded answer, and hallucinated answer [2]. White-box methods such as EGH use internal representations and gradients [3], which can be effective but require privileged model access. Ensemble and calibrated frameworks such as HALT-RAG combine retrieval, NLI, calibration, or abstention mechanisms [4], [5], improving factuality assessment at the cost of additional components and inference overhead.

Judge-based methods, including LLM-as-a-Judge [6], Self-CheckGPT [7], and MetaQA [8], evaluate hallucination through prompting, sampling, or response mutation. They are flexible but can be expensive and stochastic. Lightweight feature-based detectors rely on lexical overlap, TF-IDF, answer length, or numeric consistency [9]; these are efficient but limited for semantic reasoning. Reference-based factual consistency methods such as AlignScore [10] and MiniCheck [11] assess whether generated content is supported by a source document and are strong black-box baselines.

TABLE I: Comparison of hallucination detection approach families in terms of methodology and deployment practicality.

Approach	Supervised	Fine-tuned	Retrieval	Model Access	Deterministic	Complexity	Deployability
Feature-based Detector [9]	Yes	No	No	Black-box	Yes	Low	High
EGH [3]	Yes	Partial	No	Internal	Yes	Medium	Low
HALT-RAG [4]	Hybrid	No	Yes	Black-box	No	High	Medium
LLM-as-a-Judge [6]	No	No	Optional	Black-box	No	High	Low
SelfCheckGPT [7]	No	No	No	Black-box	No	Medium	Medium
Ours	Yes	Yes	Optional	Black-box	Yes	Low	High
Ours + Summarization	Yes	Yes	Optional	Black-box	Yes	Low	Very High

Complexity definition: Complexity denotes system and deployment overhead, considering model access requirements, number of components, inference cost, and dependence on external retrieval or sampling.

This work differs by integrating question-aware evidence compression directly into a supervised (K, Q, A) hallucination verifier. The comparison between original passages, summarized passages, and length-matched truncation isolates whether gains come from shorter inputs alone or from question-focused evidence selection. Table I summarizes the deployment trade-offs.

The key deployment distinction is that the proposed verifier performs deterministic inference over textual inputs. It does not require access to hidden states, gradients, sampling distributions, or multiple generations. This is important in practical API-based QA systems, where model internals may be inaccessible and repeated sampling may be too expensive. At the same time, the framework does not claim that summarization itself is novel; rather, it evaluates whether question-aware summarization is useful when placed before a supervised verifier and tested under controlled in-domain and long-context shifted conditions.

III. DATASET AND PROBLEM SETUP

A. HaluEval-QA and Split

HaluEval-QA pairs each question with a knowledge passage and both grounded and hallucinated answers [2]. Each original record is converted into two supervised samples: (K, Q, A_{true}) labeled as non-hallucination (1), and (K, Q, A_{hall}) labeled as hallucination (0). We use a 70/30 train-test split at the original-record level before constructing binary samples, so sibling supported and hallucinated variants never cross splits. Leakage analysis confirmed zero question overlap, zero knowledge-passage overlap, zero shared original records, and zero sibling leakage.

TABLE II: Dataset statistics used in our experiments.

Split	#Instances	#Hallucination (0)	#Non-hallucination (1)
Train	14,000	7,000	7,000
Test	6,000	3,000	3,000

The task is binary evidence-answer verification. Supported answers are grounded in the provided passage, while hallucinated answers contradict or introduce unsupported information. For example, “School Rumble” is supported when the evidence states that Unscandal performed its theme music and that the manga was written and illustrated by Jin Kobayashi; by contrast, answering “Harry” for a context stating that Joey Richter co-starred as Ron Weasley is hallucinated.

This construction keeps the verification decision explicit: the model is not asked to generate the answer, but to judge whether the candidate answer is entailed by the supplied evidence. This also makes the setup compatible with black-box QA systems,

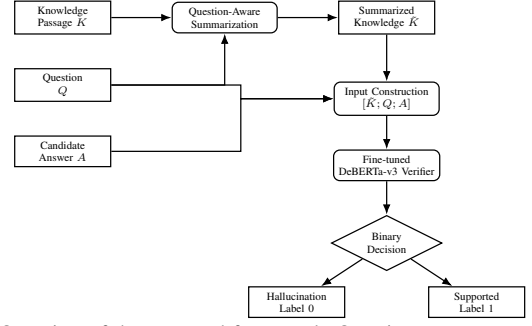


Fig. 1: Overview of the proposed framework. Question-aware summarization reduces context noise and preserves question-relevant evidence, improving hallucination detection on long knowledge passages. The summarized evidence is then verified using a lightweight supervised classifier.

because the detector only needs the retrieved or provided passage, the user question, and the generated answer.

Example 1 (Supported Answer, Label = 1). Question: What anime did Unscandal perform theme music for and is written and illustrated by Jin Kobayashi? **Answer:** School Rumble. **Knowledge excerpt:** The band performed the theme music for the anime *School Rumble*, and *School Rumble* is written and illustrated by Jin Kobayashi. **Interpretation:** The answer is directly supported by the evidence.

Example 2 (Hallucinated Answer, Label = 0). Question: Joey Richter co-starred as which Harry Potter character in 2009 fan parody musicals? **Answer:** Joey Richter co-starred as Harry. **Knowledge excerpt:** Richter co-starred as Ron Weasley in *A Very Potter Musical*. **Interpretation:** The answer contradicts the evidence because the correct character is Ron Weasley rather than Harry.

IV. METHOD

Hallucination verification assesses consistency between a candidate answer and evidence. Figure 1 shows the proposed pipeline: optionally summarize the knowledge conditioned on the question, construct the verifier input, and classify the answer as supported or hallucinated.

A. Task Definition and Input Representation

Given (K, Q, A) , the verifier predicts $y \in \{0, 1\}$, where 1 means supported and 0 means hallucinated. This formulation makes the verifier a post-generation filter: it can accept an answer from any generator and decide whether it is grounded in the available knowledge. The encoder input is

$$[\text{CLS}] \ K \ [\text{SEP}] \ Q \ [\text{SEP}] \ A \ [\text{SEP}],$$

padded or truncated to 512 tokens. This sequence allows the encoder to jointly attend to evidence, question, and answer rather than scoring the answer in isolation.

B. Verifier Architecture and Objective

We fine-tune DeBERTa-v3-base; the final $[\text{CLS}]$ representation is passed to a linear classifier, $z = Wh_{[\text{CLS}]} + b$,

with $p(y \mid K, Q, A) = \text{softmax}(z)$. Training minimizes cross-entropy over the binary labels,

$$\mathcal{L} = - \sum_{c \in \{0,1\}} \mathbb{I}[y = c] \log p(y = c \mid K, Q, A).$$

Compared with judge-based or sampling-based methods, this yields one stable decision per input. We also test question-unaware evidence compression by replacing K with a generic summary $\hat{K}$ while keeping the verifier unchanged, allowing compression alone to be separated from question-aware evidence selection.

C. Why Question-Aware Compression Helps

The central intuition is that hallucination verification is not only a classifier-design problem; it is also an evidence-representation problem. If the verifier receives a long passage with many unrelated entities, dates, and events, the relevant support signal may be diluted. Question-aware summarization acts as an evidence-selection step that keeps the parts most likely to answer the question while removing distractors. This is especially important for long-context settings, where the answer may depend on a small subset of the passage.

The length-matched truncation experiment is therefore important. If summarization helped only because the input became shorter, truncating HotpotQA passages to the same average length should have produced similar performance. Instead, truncation reaches only 0.5457 macro-F1, while question-aware summarization reaches 0.7905. This gap shows that the model benefits from relevance-aware compression rather than compression alone.

V. KNOWLEDGE SUMMARIZATION FOR DETECTION

Knowledge passages often contain information unrelated to the question. Question-aware summarization compresses K into $\hat{K}$ using the prompt: “Summarize the following knowledge to answer the question. Question: Q . Knowledge: K .” This can improve the signal-to-noise ratio but may also omit subtle evidence, so its effect is evaluated empirically. We compare BART, FLAN-T5, PEGASUS, and generic summarization. The verifier remains unchanged and receives $(\hat{K}, Q, A)$, isolating the effect of evidence preprocessing.

The motivation is that the relevant evidence for a specific question may occupy only a small portion of the full passage. If irrelevant sentences dominate the input, the verifier may attend to misleading entities, dates, or locations. However, summarization can introduce its own risk by dropping the answer span or compressing a reasoning chain too aggressively. For this reason, the paper evaluates summary faithfulness, answer preservation, error concentration, and a length-matched HotpotQA control rather than assuming that shorter context is always better.

VI. EXPERIMENTAL SETUP AND RESULTS

All experiments use the HuggingFace Transformers pipeline with AdamW optimization. Accuracy, precision, recall, and F1 are computed on the 6,000-sample HaluEval-QA test set, treating the supported class as positive. We include AlignScore

TABLE III: Verification model training hyperparameters.

Parameter	Value
Backbone	DeBERTa-v3-base
Max sequence length	512
Optimizer	AdamW
Learning rate	2×10^{-5}
Weight decay	0.01
Warmup ratio	0.05
Train batch size	8
Eval batch size	8
Epochs	2
Split	70/30 stratified

and MiniCheck as black-box factual-consistency baselines, lexical exact match as a diagnostic heuristic, and zero-shot NLI DeBERTa to isolate the value of task-specific fine-tuning.

We evaluate the verifier with standard binary classification metrics. Let TP , TN , FP , and FN denote true positives, true negatives, false positives, and false negatives, respectively. Accuracy measures overall correctness,

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN},$$

precision measures how many predicted supported answers are actually supported,

$$Precision = \frac{TP}{TP + FP},$$

recall measures how many truly supported answers are recovered,

$$Recall = \frac{TP}{TP + FN},$$

and F1 is the harmonic mean of precision and recall,

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}.$$

These metrics jointly evaluate overall correctness and class-specific verification quality. For the HotpotQA-derived benchmark, macro-F1 is reported because the experiment is used as a balanced out-of-distribution verification stress test.

The hyperparameters in Table III follow common transformer fine-tuning practice and preliminary validation. Training used an NVIDIA Tesla T4 GPU, seed 42, the same split, and two epochs, requiring about 1.5 hours. The learning rate of 2×10^{-5} provided stable optimization without large validation fluctuations, while the batch size of 8 was selected due to GPU memory limits at 512 tokens. A warmup ratio of 0.05 and weight decay of 0.01 were used to improve stability and reduce overfitting. The two-epoch setting was kept because validation performance converged rapidly and additional training was not necessary.

TABLE IV: Training convergence of the DeBERTa verifier.

Epoch	Training Loss	Validation Loss	Accuracy	Precision	F1
1	0.4414	0.4057	0.9760	0.9612	0.9764
2	0.4183	0.3940	0.9770	0.9640	0.9773

TABLE V: Performance stability across three random seeds on the FLAN-T5 summarized knowledge setting.

TABLE VI: Comparison between zero-shot NLI and task-specific fine-tuned verification on the HaluEval-QA test set.

Method	Accuracy	Precision	Recall	F1
Zero-shot NLI DeBERTa	0.5280	0.6042	0.1623	0.2559
Fine-tuned DeBERTa	0.9770	0.9640	0.9910	0.9773
Ours + FLAN-T5	0.9848	0.9815	0.9883	0.9849

TABLE VII: Impact of different summarization strategies on downstream hallucination detection.

Summarization Strategy	Accuracy	Precision	Recall	F1
Original (No summarization)	0.9770	0.9640	0.9910	0.9773
Overall Summarization (Generic)	0.9767	0.9631	0.9913	0.9770
Question-Aware BART-CNN	0.9790	0.9666	0.9923	0.9793
Question-Aware PEGASUS	0.9782	0.9656	0.9917	0.9785
Question-Aware FLAN-T5	0.9848	0.9815	0.9883	0.9849

TABLE VIII: Faithfulness validation of the FLAN-T5 summaries used in the proposed framework.

Metric	Value
Summary Faithfulness (MiniCheck)	92.98%
Answer Preservation Rate	74.78%
Compression Ratio	0.389
Context Reduction	61.05%

Seed	Accuracy	Precision	Recall	F1
42	0.9848	0.9815	0.9883	0.9849
123	0.9787	0.9674	0.9907	0.9789
999	0.9787	0.9656	0.9927	0.9790
Mean	0.9807	0.9715	0.9906	0.9809
Std	0.0035	0.0087	0.0021	0.0035

Tables IV and V show rapid convergence and low seed variance. The average F1 across seeds is 0.9809 with standard deviation 0.0035, close to the main seed-42 result.

Table VI shows that zero-shot NLI is far below the fine-tuned verifier, confirming that supervised optimization is essential. Table VII shows that generic summarization is not enough, while question-aware FLAN-T5 gives the best HaluEval-QA F1 of 0.9849. Representative outputs further confirmed correct supported and hallucinated predictions, such as verifying “ECCO” as supported and detecting the unsupported Stephen Sondheim answer for “Tick, Tick...Boom!”.

The small difference between the original and summarized settings on HaluEval-QA should be interpreted carefully. The original verifier is already near saturation on this controlled benchmark, so the improvement is numerically modest. Nevertheless, the summarized setting improves precision while maintaining high recall, suggesting that question-focused evidence helps reduce unsupported supported-label predictions.

Example Output. For the question about Hanni Toosbuy Kasprzak and the Danish shoe manufacturer founded in 1963, the answer “ECCO” is correctly predicted as supported. For the question about the Pulitzer-prize-winning composer who wrote *Tick, Tick...Boom!*, the answer attributing the work to Stephen Sondheim is correctly detected as hallucinated.

The FLAN-T5 summaries are mostly faithful to the source evidence (Table VIII) while reducing context length by 61.05%. Faithfulness and answer preservation differ because a summary may remain factually supported while omitting the exact answer span. McNemar’s test confirms the summarized

improvement is statistically significant ($\chi^2 = 21.37$, exact $p = 2.48 \times 10^{-6}$).

A. Summary of Controlled Comparisons

The study uses several controls to avoid over-interpreting a single result. The no-summarization setting measures the ability of the verifier to classify answers using the full evidence passage. The generic summarization setting tests whether compression without question conditioning is sufficient. The BART, PEGASUS, and FLAN-T5 settings compare different question-aware summarizers under the same verifier. The zero-shot NLI baseline tests whether a pretrained entailment model can solve the task without fine-tuning. The external baselines test whether general factual consistency metrics transfer to this binary QA verification setup. Finally, the HotpotQA length-matched control tests whether the long-context gain is caused by relevance-aware evidence selection rather than by shorter input length alone.

Together, these comparisons support the same conclusion from different angles: the verifier needs both task-specific supervision and well-focused evidence. Fine-tuning alone works well in controlled HaluEval-QA, but it becomes weak under long-context distribution shift. Shortening alone helps slightly, but question-aware compression gives a much larger improvement. This is why the proposed framework combines a simple supervised verifier with optional question-aware preprocessing instead of relying only on a more complex verifier architecture.

B. Reproducibility Notes

All reported HaluEval-QA experiments use the same held-out 6,000-sample test set, the same train-test split strategy, and the same hyperparameter configuration. The seed-stability experiment repeats the summarized FLAN-T5 setting with seeds 42, 123, and 999 to check whether the gain depends on random initialization. The low standard deviation across seeds indicates that the result is stable. Leakage checks are also important because each original HaluEval-QA record produces a supported and a hallucinated sample; splitting after sample construction could allow related sibling examples to appear across train and test. Splitting at the original-record level prevents this issue.

C. Detailed Result Interpretation

The HaluEval-QA results show that the supervised verifier already learns a strong evidence-answer consistency function from original passages. This is expected because the dataset is balanced, controlled, and built from paired supported and hallucinated answers. However, this controlled structure also means that HaluEval-QA alone is not sufficient to prove broad real-world generalization. For this reason, the paper treats HaluEval-QA mainly as an in-distribution validation setting and relies on the HotpotQA-derived benchmark as the more informative stress test.

Question-aware summarization changes the verification problem by presenting the detector with a smaller and more focused evidence passage. The improvement over generic

TABLE IX: Ablation study of the proposed framework on HaluEval-QA.

Configuration	Accuracy	F1
DeBERTa Verifier Only	0.9770	0.9773
+ Generic Summarization	0.9767	0.9770
+ Question-Aware BART	0.9790	0.9793
+ Question-Aware PEGASUS	0.9782	0.9785
+ Question-Aware FLAN-T5	0.9848	0.9849

TABLE X: Comparison with external hallucination detection baselines on the HaluEval-QA test set.

Method	Accuracy	Precision	Recall	F1
Zero-shot NLI DeBERTa	0.5280	0.6042	0.1623	0.2559
Lexical Exact Match (Original)	0.8527	0.9962	0.7080	0.8277
Lexical Exact Match (Summarized)	0.8822	0.9948	0.7683	0.8670
MiniCheck [11] (Original)	0.7378	0.7381	0.7378	0.7377
MiniCheck [11] (Summarized)	0.7400	0.7487	0.7400	0.7377
AlignScore [10] (Original)	0.7708	0.7227	0.8790	0.7932
AlignScore [10] (Summarized)	0.7928	0.7777	0.8198	0.7982
Ours (Original)	0.9770	0.9640	0.9910	0.9773
Ours + FLAN-T5	0.9848	0.9815	0.9883	0.9849

TABLE XI: Comparison of fine-tuned transformer verifiers using question-aware FLAN-T5 summarized knowledge.

Model	Accuracy	Precision	Recall	F1
BERT-base	0.9322	0.9602	0.9017	0.9300
ELECTRA-base	0.9412	0.9607	0.9200	0.9399
RoBERTa-base	0.9625	0.9636	0.9613	0.9625
DeBERTa-v3-base (Ours)	0.9848	0.9815	0.9883	0.9849

summarization indicates that not all compression is equally useful: removing irrelevant context without conditioning on the question may shorten the passage but does not necessarily preserve the specific evidence needed to judge the answer. FLAN-T5 performs best in our experiments because its summaries preserve more question-relevant evidence for the downstream verifier.

The statistical test, faithfulness analysis, and answer-preservation analysis are included to make this interpretation more reliable. McNemar’s test checks that the summarized improvement is not due to random disagreement between two prediction sets. MiniCheck-based faithfulness evaluates whether summaries remain supported by the original knowledge. Answer preservation then examines whether the explicit answer span survives summarization, which is important because many errors concentrate when the answer span is dropped.

The ablation in Table IX confirms that the gain comes mainly from question-focused evidence rather than compression alone. Table X shows that lexical matching and factual-consistency scorers benefit from summarized knowledge, but the supervised DeBERTa verifier remains substantially stronger.

Lexical exact match is included because HaluEval-QA contains many short factoid answers, making surface overlap a useful diagnostic signal. However, the gap between lexical matching and the supervised verifier shows that semantic evidence-answer verification is still needed. AlignScore is the strongest external factual-consistency baseline, but it remains below the fine-tuned verifier, indicating that task-specific supervised training on (K, Q, A) triples is more effective for this binary setting.

Table XI shows that the framework is not limited to one

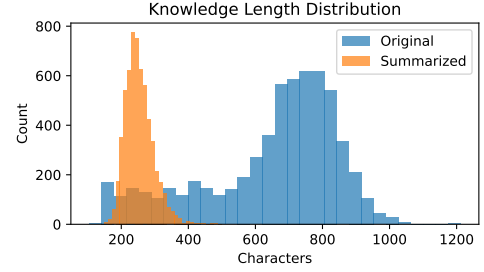


Fig. 2: Knowledge length distribution before and after summarization.

TABLE XII: Inference efficiency comparison between original and summarized knowledge.

Setting	Avg Tokens	Latency (ms)	Throughput (samples/s)	GPU Time (s)	GPU Mem. (GB)
Original	110.95	15.99	62.56	95.91	2.68
Summarized	63.25	9.71	103.02	58.24	1.59

encoder, although DeBERTa-v3 with FLAN-T5 summarization gives the strongest verification performance.

D. Baseline and Backbone Analysis

The external baseline results provide two complementary observations. First, lexical exact match is a useful but incomplete signal. It performs reasonably well on HaluEval-QA because many answers are short factoid spans, yet it cannot reliably capture paraphrased support, implicit evidence, or contradictions that require semantic interpretation. Second, AlignScore and MiniCheck are strong general-purpose factual-consistency tools, but they are not optimized specifically for this paired (K, Q, A) binary verification task. The fine-tuned verifier benefits from direct supervision on the same input structure used at test time.

The backbone comparison further shows that the proposed idea is not tied to DeBERTa alone. BERT, ELECTRA, and RoBERTa all achieve strong performance when trained on the summarized setting, which supports the usefulness of the task formulation and preprocessing. Nevertheless, DeBERTa-v3 performs best, likely because its disentangled attention and enhanced position modeling help represent the relationship among evidence, question, and candidate answer under limited input length.

E. Analysis, Efficiency, and Errors

Figure 2 shows that summarization reduces knowledge length, and Figure 3 shows that it mainly reduces false positives. The length histogram indicates a clear shift toward shorter summarized inputs, while the confusion matrices show that the summarized setting keeps both classes highly accurate. This supports the idea that question-aware summarization removes distractor context without eliminating most evidence needed for verification.

Verifier-side summarization reduces average tokens from 110.95 to 63.25, latency by about 39%, GPU time from 95.91 s to 58.24 s, and memory from 2.68 GB to 1.59 GB (Table XII). Table XIII clarifies that online summarization is not cost-free. The standalone FLAN-T5 summarizer is a text-generation stage, not a hallucination classifier, so accuracy and

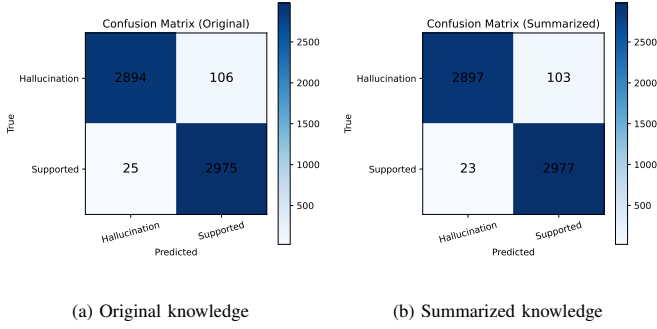


Fig. 3: Confusion matrices using original and summarized knowledge passages on the HaluEval-QA dataset.

TABLE XIII: End-to-end latency analysis including the FLAN-T5 summarizer.

Setting	KQ Rec.	KQA Checks	Summ. Tok.	Verif. Tok.	Latency (ms)	Throughput (samples/s)
Original detector only	N/A	6000	N/A	113.92	4.01	249.67
Cached summarized detector only	N/A	6000	N/A	75.41	3.28	304.97
Generated summarized detector only	N/A	6000	N/A	97.47	4.05	246.82
FLAN-T5 summarizer only	3000	N/A	132.81	N/A	264.22	3.78
Online end-to-end pipeline	3000	6000	132.81	97.47	268.27	3.73
Benchmark end-to-end (summary reused)	3000	6000	132.81	97.47	136.16	7.34

N/A indicates that the metric is not applicable to that stage.

TABLE XIV: Categorization of remaining verification errors on the HaluEval-QA test set.

Error Category	Count	Percentage
Temporal Reasoning	32	23.19%
Location Confusion	29	21.01%
Entity Confusion	25	18.12%
Multi-hop / Comparative Reasoning	21	15.22%
Subtle Evidence Mismatch	19	13.77%
Numerical Reasoning	7	5.07%
Lexical Ambiguity	5	3.62%

TABLE XV: Error concentration according to answer preservation in FLAN-T5 summarized knowledge.

Summary Group	Total Samples	Errors	Error Rate	Share of All Errors
Answer Dropped	1606	29	1.81%	61.70%
Answer Preserved	4394	18	0.41%	38.30%

F1 are not defined for it. The online end-to-end latency is computed as one FLAN-T5 summarization pass for a (K, Q) pair plus one DeBERTa verification pass for one candidate answer. The benchmark end-to-end row amortizes the same generated summary across the two candidate answers derived from each original record. FLAN-T5 dominates latency, so the summarized pipeline is most efficient when summaries are cached, generated offline, or reused.

Error analysis (Table XIV) shows that remaining failures mainly involve temporal reasoning, location confusion, entity confusion, multi-hop reasoning, and subtle mismatches. Table XV shows that dropped answer spans account for 61.70% of all remaining errors despite being 26.77% of samples, but answer-preserved cases still produce errors, so failures are not explained by answer dropping alone.

Representative errors illustrate these categories. In temporal reasoning, the verifier incorrectly accepted “March 1994” for a release-date question involving The Prodigy’s single *Poison*, where multiple dates made the evidence difficult to separate. In entity confusion, the verifier incorrectly rejected

TABLE XVI: Length distribution and verification performance across HaluEval and HotpotQA settings.

Setting	Total	Avg Tokens	Accuracy	Reported F1
HaluEval Original knowledge	6000	110.95	0.9770	0.9773
HaluEval Question-aware FLAN-T5 summarization	6000	63.25	0.9848	0.9849
HotpotQA Original knowledge	1978	1186.30	0.5040	0.5040
HotpotQA Length-matched truncation	1978	81.94	0.5966	0.5457
HotpotQA Question-aware FLAN-T5 summarization	1978	81.95	0.7917	0.7905

HaluEval reports the supported-class F1 used in the in-distribution evaluation, while HotpotQA reports macro-F1 because it is used as an out-of-distribution balanced verification benchmark.

TABLE XVII: Generalization results with a length-matched control on the HotpotQA-derived binary hallucination-detection benchmark.

Setting	Total	Avg Tokens	Accuracy	Macro Precision	Macro Recall	Macro F1
Lexical Exact Match (Original)	1978	1186.30	0.6571	0.9670	0.6571	0.4868
Lexical Exact Match (Summarized)	1978	81.95	0.7768	0.9507	0.7768	0.7234
Original knowledge	1978	1186.30	0.5040	0.5040	0.5040	0.5040
Length-matched truncation	1978	81.94	0.5966	0.6749	0.5966	0.5457
Question-aware FLAN-T5 summarization	1978	81.95	0.7917	0.7988	0.7917	0.7905

the supported answer “University of Manchester” when several scientists and institutions appeared in the context. In multi-hop reasoning, the verifier incorrectly accepted “OK Computer” for a question requiring aggregation of producer, engineer, and artwork evidence across multiple facts. These examples show that the remaining errors are concentrated in fine-grained reasoning rather than simple direct matching.

F. Representative Failure Patterns

The remaining errors are not random. Temporal reasoning failures usually happen when several dates or event periods appear in the same passage and the candidate answer is close to, but not exactly, the supported date. Location and entity confusion occur when the passage mentions multiple similar people, organizations, or places and the verifier attaches a property to the wrong entity. Multi-hop errors appear when the evidence must be combined across several facts, which is more difficult than direct sentence-level entailment.

These patterns are important for deployment because they suggest where abstention or confidence thresholds may be most useful. In applications where dates, names, or legal/medical details are critical, a hallucination detector should not only output a binary label but may also need to trigger human review or retrieval refinement when confidence is low. The current framework can support such extensions because it already produces class probabilities from the verifier head.

G. HotpotQA-Derived Generalization

We evaluate a balanced HotpotQA-derived binary benchmark not used in training. Each HotpotQA example becomes a supported sample using the ground-truth answer and a hallucinated sample using type-aware negative answer substitution, with duplicate and trivial replacements filtered. The benchmark contains 1,978 samples (989 supported and 989 hallucinated). It is a controlled synthetic verification benchmark, not a complete collection of naturally generated LLM hallucinations. Because HotpotQA passages are long and multi-hop, it tests distribution shift and long-context reasoning. A length-matched truncation control separates shorter input length from question-focused evidence selection.

Tables XVI and XVII show a major length shift: HotpotQA original passages average 1186.30 tokens. The original verifier

is near chance (0.5040 macro-F1). Length-matched truncation improves only to 0.5457, while question-aware FLAN-T5 summarization reaches 0.7905 with nearly the same average length. Therefore, the gain is not explained by shortening alone; it comes mainly from selecting question-relevant evidence. Lexical exact match also improves after summarization, but remains below the supervised summarized verifier.

The balanced construction means macro recall closely matches accuracy, avoiding class-prior bias. The lexical baseline performs weakly on original long HotpotQA passages because the answer may appear among many irrelevant facts, but summarization improves it by making answer matching easier. Still, the supervised summarized verifier achieves the strongest result, indicating that the benefit extends beyond surface overlap. This experiment is the strongest evidence for the central claim because HaluEval-QA is already near-saturated, whereas HotpotQA exposes the verifier to much longer and more reasoning-intensive contexts.

VII. DISCUSSION AND PRACTICAL DEPLOYMENT

The results should be interpreted differently across benchmarks. HaluEval-QA validates that the detector learns evidence-answer consistency in a controlled balanced setting, but its high F1 may partly reflect cleaner paired construction and stronger lexical cues. HotpotQA-derived evaluation is a stronger long-context stress test because the passages are longer, more multi-hop, and distributionally different from training data. The length-matched control confirms that question-aware summarization contributes more than input shortening.

At the same time, the HotpotQA-derived benchmark uses synthetic hallucinated answers generated through type-aware answer substitution. Therefore, the out-of-distribution result should be interpreted as evidence for improved long-context binary verification, not as definitive proof of robustness to every naturally occurring LLM hallucination. Broader evaluation should include naturally generated hallucinations, noisier retrieval outputs, and multi-document evidence settings.

The proposed verifier is deterministic, black-box deployable, and does not require model internals, retrieval modules, ensembles, or sampling. When summarization is disabled, the system is a single lightweight verifier. When summarization is enabled, it should be described as a lightweight verifier with an optional preprocessing stage, not a cost-free end-to-end system. In practice, summaries can be generated offline, cached, or reused across multiple candidate answers, making the method suitable for long-context QA pipelines that need a practical balance of performance, efficiency, and reproducibility. The deployment tradeoff is therefore clear: summarization adds preprocessing cost when used online, but improves robustness and reduces verifier-side computation when summaries are cached or reused.

A. Practical Use in QA Systems

A practical QA system can use the verifier in several ways. If the answer is predicted as supported with high confidence, the system may return it normally. If the answer is predicted

as hallucinated, the system may reject the answer, ask the generator to regenerate using the same evidence, retrieve additional evidence, or abstain. When the evidence passage is long, question-aware summarization can be applied before verification to reduce irrelevant context. This design keeps the verifier independent from the generator and therefore compatible with black-box LLMs, retrieval-augmented systems, or human-written candidate answers.

The most efficient deployment scenario is offline or cached summarization. In many QA pipelines, the same question and retrieved passage may be checked against multiple candidate answers, regenerated answers, or alternative model outputs. In that case, one summary can be reused for several verification calls, amortizing the summarization cost while preserving the verifier-side speedup. When summaries must be generated online for every query, the system trades additional preprocessing latency for improved robustness on long and noisy contexts.

B. Limitations

Several limitations should be noted. First, the HaluEval-QA test set is balanced and controlled, while real QA systems may face unbalanced hallucination rates, noisy retrieval, ambiguous questions, or adversarial answers. Second, the HotpotQA-derived benchmark uses type-aware answer substitution to generate negative examples. This creates a useful controlled binary benchmark, but the hallucinated answers may differ from naturally generated LLM errors in fluency and plausibility. Third, summarization may remove exact answer spans or compress multi-hop evidence too aggressively, which explains why answer preservation is analyzed separately. Finally, although the verifier is lightweight, online summarization introduces a real preprocessing cost.

These limitations do not weaken the main claim; rather, they define its scope. The results support the claim that question-aware evidence compression can improve supervised hallucination verification, especially under long-context distribution shift. They do not claim universal robustness across every hallucination type or deployment environment.

VIII. CONCLUSION

We presented a supervised hallucination detection framework for knowledge-grounded QA by fine-tuning DeBERTa-v3-base over (K, Q, A) triples and optionally applying question-aware knowledge summarization. The verifier achieves strong HaluEval-QA performance, and FLAN-T5 summarization improves F1 to 0.9849. Under HotpotQA-derived long-context distribution shift, question-aware summarization improves macro-F1 from 0.5040 to 0.7905, while a length-matched truncation control reaches only 0.5457. These results show that question-aware evidence compression improves evidence-answer alignment, verifier-side efficiency, and long-context robustness.

The generalization claim should be understood as evidence on the HotpotQA-derived synthetic binary benchmark evaluated here, not definitive proof across all naturally occurring LLM hallucinations. Future work will evaluate additional out-of-distribution hallucination benchmarks, naturally generated

errors, noisy retrieval, multi-document evidence, calibration, and abstention strategies.

A further extension is to connect the verifier with adaptive retrieval and answer regeneration. Instead of only accepting or rejecting an answer, the system could identify low-confidence cases, retrieve more targeted evidence, regenerate the answer, or abstain when evidence remains insufficient. Another direction is to improve summarization with explicit answer-preservation constraints so that summaries remain concise while retaining the facts needed for verification. These extensions follow directly from the current findings: the detector benefits most when the evidence is both compact and question-relevant.

REFERENCES

- [1] E. E.-H. Shemis, A. M. Gadallah, and H. A. Hefny, “A data-sensitive approach for fuzzy concept extraction,” *International Journal of Intelligent Engineering & Systems*, vol. 11, no. 5, 2018.
- [2] J. Li, X. Cheng, W. X. Zhao, J.-Y. Nie, and J.-R. Wen, “Halueval: A large-scale hallucination evaluation benchmark for large language models,” in *Proc. EMNLP*, 2023, pp. 6449–6464.
- [3] L. Zhang et al., “Embedding and gradient: A white-box method for detecting hallucinations in large language models,” in *Proc. EMNLP*, 2024.
- [4] S. Goswami and S. Kurra, “Halt-rag: A task-adaptable framework for hallucination detection with calibrated nli ensembles and abstention,” 2025.
- [5] C. Ceccarelli, A. Raganato, and M. Viviani, “Knowledge-grounded detection of factual hallucinations in large language models,” in *Proc. CLiC-it*, 2025, pp. 195–201.
- [6] H. Huang et al., “An empirical study of llm-as-a-judge for llm evaluation: Fine-tuned judge model is not a general substitute for gpt-4,” in *Findings of ACL*, 2025.
- [7] P. Manakul, A. Liusie, and M. Gales, “Selfcheckgpt: Zero-resource black-box hallucination detection for generative large language models,” in *Proc. EMNLP*, 2023.
- [8] B. Yang, M. A. Al Mamun, J. M. Zhang, and G. Uddin, “Hallucination detection in large language models with metamorphic relations,” *Proc. ACM on Software Engineering*, vol. 2, no. FSE, 2025.
- [9] T. Reed and G. Mason, “Hallucination detection and confidence calibration for large language model outputs: Reproducible experiments on halueval,” *Artificial Intelligence and Machine Learning Review*, vol. 6, no. 4, pp. 1–17, 2025.
- [10] Y. Zha, Y. Yang, R. Li, and Z. Hu, “Alignscore: Evaluating factual consistency with a unified alignment function,” in *Proc. ACL*, 2023, pp. 11328–11348.
- [11] L. Tang, P. Labal, and G. Durrett, “Minicheck: Efficient fact-checking of llms on grounding documents,” in *Proc. EMNLP*, 2024.